

Deepak Mallampati

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Kent, OH - US-based; open to UK relocation with sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer who builds accuracy-critical LLM systems for regulated environments. Production track record from healthcare AI: Retrieval-Augmented Generation with **~35% precision gain** and **>90% grounding alignment**, agentic LLM workflows with **>30% hallucination reduction** via schema-validated agent contracts, and FastAPI/Docker services with **~30% post-deployment defect reduction**. Full-stack capability across React/TypeScript front ends and Python/FastAPI back ends; multi-LLM product shipped to Chrome Web Store. Master's in Computer Science (Kent State). Comfortable owning real-time AI workflows end-to-end where control, accuracy, and trustworthiness are non-negotiable.

CORE COMPETENCIES

LLM Agents (Accuracy-Critical) • RAG & Grounding • Embeddings & Vector Search • Schema-Validated Agent Contracts • Evaluation Pipelines • Real-time AI Workflows • Full-stack (React/TS + FastAPI) • Python • TypeScript • Docker • HIPAA-Conscious Deployment

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present | USA

Jr. AI/ML Engineer Trainee - Healthcare Technology (HIPAA-conscious)

- Built **Retrieval-Augmented Generation** for clinical knowledge retrieval; recursive semantic chunking + transformer embeddings improved precision **~35%**, retrieval-grounded response alignment exceeded **90%**.
- Developed **agentic LLM workflows** for multi-step healthcare queries with structured reasoning, tool discipline, and grounding rules; improved response stability **~25%** and reduced hallucinations **>30%**.
- Packaged ML/LLM inference as **FastAPI/Flask** services, containerized with **Docker**, with structured logging, eval audit trails, and load simulation; reduced post-deployment defects **~30%**.
- Owned predictive ML pipelines (scikit-learn, XGBoost); improved recall **15-20%** on high-risk categories while holding precision **>90%**.
- Maintained HIPAA-conscious data governance: de-identification, lineage docs, evaluation audit trails, system-limitation documentation for stakeholders.

Suvidha Foundation

Jan 2022 - Mar 2022 | India

Machine Learning Engineer - Applied NLP, Video Analytics

- Transformer-based summarization across 5,000+ sessions; hierarchical long-context summarization cut manual review time **60-70%**.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs **~30%**, raised answer accuracy **>85%**.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023 | India

Database & DevOps Performance Engineer (Intern) - Compliance-sensitive enterprise ERP

- Built **CI/CD pipelines (Jenkins)** for version-controlled SQL and stored-procedure deploys; reduced deployment errors **>30%**, release cycle time **~35-40%**.
- Supported RBAC implementation and audit logging for financial modules in a compliance-sensitive environment.

PROJECTS

Agentic Healthcare Claims & Fraud Risk Intelligence System

Multi-agent pipeline (Intake → Validation → Consistency → Duplicate Detection → Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations; RAG-grounded CPT/ICD validation against vector-indexed policy docs, ANN duplicate detection, and explainable audit-ready reasoning traces - the architectural pattern for accuracy-critical agent systems.

Multi-agent LLM, schema contracts, RAG, vector search, evaluation, audit trails

Manga Lens - Real-Time Multi-LLM Vision Pipeline (shipped, Chrome Web Store)

Full-stack browser extension shipped solo - integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v) behind a common interface, with per-provider payload handling, 7-day translation cache, and per-domain selector configs for 29 sites. Demonstrates real-time AI workflow design and production multi-LLM orchestration.

TypeScript, Manifest V3, multi-LLM orchestration, real-time vision pipelines

Dream Decoder - Multimodal AI Platform

Full-stack **FastAPI + React/TypeScript/Vite/Tailwind** app coordinating multimodal APIs. Introduced intermediate **structured prompt transformation layers** - improved contextual alignment **~30%** and first-pass image success **~25-30%** vs naïve direct-prompt orchestration.

React, TypeScript, FastAPI, structured prompt orchestration

Driver Drowsiness Detection - Real-Time YOLOv8

Replaced a two-stage CNN pipeline with a unified **YOLOv8** detection-and-classification model; reduced inference latency **~30%**; sliding-window confidence aggregation suppressed blink-driven false positives **~25%**.

YOLOv8, OpenCV, real-time inference

EDUCATION

Master's Degree, Computer Science - Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

LLM & GenAI: Agentic workflows, RAG, structured outputs, prompt engineering, evaluation pipelines, grounding, embeddings, vector/semantic search, responsible AI

Languages & Frameworks: Python, FastAPI, Flask, TypeScript, React, Node.js (adjacent), SQL (PostgreSQL, T-SQL, SQLite)

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, LangChain, LlamaIndex

Infra & DevOps: Docker, Jenkins, CI/CD, RESTful APIs, observability/logging, Grafana

Domains: Healthcare AI (HIPAA-conscious), accuracy-critical AI, enterprise AI agents, real-time multi-LLM systems